One key piece of data is missing from here that we've been depending on from every other data source we've looked at so far; did you notice it? Dates are missing! That may seem to put a kink in things if you're talking about collecting data and mapping it over time, but don't worry; if you dig deeply enough you'll find them. Jenkins assumes that when you hit a URL you know the current date and time. If you're going to take this data and save it somewhere, you can simply schedule the data collection at the frequency at which you want to collect data and add the dates to the database yourself. Additionally, if you look for something that references a specific time, you can get the data from that if you want. The next listing shows data from a specific build that has a date/time in the response.

Listing 5.2. Jenkins response from a specific build

```
{
    "actions": [
        {},
        {
            "causes": [
                {
                    "shortDescription": "[URLTrigger] A change within the
                     response URL invocation (<a href=\"triggerCauseAction\
                     \">log</a>)"
                }
            ]
        },
        {},
        {
            "buildsByBranchName": {
                "origin/master": {
                    "buildNumber": 1974,
                    "buildResult": null,
                    "marked": {
                        "SHA1": "54ad73e1adb22fd84fdd1dfb5c28175f743d1960",
                        "branch": [
                            {
                                "SHA1":
 "54ad73e1adb22fd84fdd1dfb5c28175f743d1960",
                                "name": "origin/master"
                            }
                        ]
                    },
```

An array of actions from this build

Details about what code is getting built